

THE NONLINEAR SCHRÖDINGER EQUATION WITH RANDOM INITIAL DATA

JAMES BONA-LANDRY

ABSTRACT. In this talk we investigate the periodic nonlinear Schrödinger equation (NLS) from a probabilistic perspective. We address some fundamental questions such as how the introduction of randomness impacts the scaling paradigm of the equation, and how this randomness is propagated by the flow. We furthermore discuss almost sure local well-posedness results in the probabilistically subcritical regime and under what conditions one can construct global-in-time solutions.

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1. INTRODUCTION & MOTIVATION

Recall that the periodic nonlinear Schrödinger equation is the dispersive semilinear PDE for a complex-valued function u on $\mathbb{R} \times \mathbb{T}^d$ given by

$$(NLS) \quad \begin{cases} iu_t + \Delta u = |u|^{p-1}u \\ u(0, x) = u_0(x) \in H^s(\mathbb{T}^d) \end{cases}, \quad p \in 2\mathbb{N} + 1, s \in \mathbb{R}.$$

We've seen a relatively complete picture of the well-posedness theory for (NLS) in terms of the regularity of the initial data. Specifically, in the subcritical regime one can always construct local-in-time solutions (via Banach fixed point arguments), and for sufficiently small initial data we can do even better and construct global-in-time solutions. On the other hand, in the supercritical regime, one can come up with pathological examples leading to ill-posedness.

The goal of this talk is to improve this picture by better understanding the supercritical regime near the criticality threshold. Specifically, although there exist examples leading to ill-posedness in this case, a natural question to ask is: what happens “generically”? In other words, what happens if we *randomize* the initial data and then study the resulting problem probabilistically? Do we have (local) well-posedness with high probability? Better yet, is it possible to construct *global*-in-time solutions?

It has long been known (particularly in the parabolic setting) that PDEs with randomness generally behave better in terms of local well-posedness. This phenomenon can largely be attributed to the effect of *probabilistic smoothing*, which comes from fact that independent random variables tend to have better linear and nonlinear estimates than arbitrary functions of the same regularity. In the '90s, Bourgain made groundbreaking strides in extending this trend to the dispersive setting by showing that in certain deterministically supercritical cases, random initial data leads to the almost sure existence of strong solutions. This work motivated some fundamental questions that have largely driven the study of the problem since:

- What is the optimal regularity exponent s for well-posedness to hold? In other words, what is the *probabilistic* criticality threshold?
- How is randomness *propagated* by the flow of (NLS)? In other words, given the initial distribution of the solution, what does the distribution look like at time t ?

The focus of this talk will be on highlighting recent work of Deng, Nahmod, and Yue [DNY20; DNY24] that answers these questions.

2. REVIEW OF PROBABILITY THEORY

2.1. Basics. A *probability space* is a measure space in which the underlying measure μ has total mass equal to one, that is, $\mu(M) = 1$. To emphasize a probabilistic viewpoint, we usually denote such a space by $(\Omega, \mathcal{F}, \mathbb{P})$, thinking of $\mathcal{F}$ as consisting of all possible “outcomes” in the sample space Ω and of $\mathbb{P}$ as prescribing to each outcome $A \in \mathcal{F}$ a probability $\mathbb{P}(A) \in [0, 1]$ of occurring.

Given another measurable space $(E, \mathcal{E})$, a *random variable* on E is simply a measurable function X from a probability space into E . In many cases, the space $(E, \mathcal{E})$ is $\mathbb{R}^n$ with the Borel σ -field, but it doesn’t need to be; one can consider random objects of virtually any kind. The requirement that X be measurable essentially ensures that one can “conduct experiments” on X (and for this reason it is a very natural one). Precisely, it means that for any $A \in \mathcal{E}$, the value $\mathbb{P}(X^{-1}(A))$, usually denoted $\mathbb{P}(X \in A)$, is defined. In other words, one can always ask the question: what is the probability that X belongs to A ?

The *law* (or *distribution*) of X , usually denoted $\mathbb{P}_X$, is the probability measure on $(E, \mathcal{E})$ given by the pushforward of $\mathbb{P}$ via X :

$$\mathbb{P}_X(A) := \mathbb{P}(X^{-1}(A)) = \mathbb{P}(X \in A), \quad A \in \mathcal{E}.$$

Sometimes it is the case that $\mathbb{P}_X$ is absolutely continuous with respect to some reference measure ν on $(E, \mathcal{E})$. In this case, the Radon-Nikodym theorem guarantees the existence of a ν -density f for $\mathbb{P}_X$ and we can characterize the distribution of X in terms of this density:

$$\mathbb{P}_X(A) = \mathbb{P}(X \in A) = \int_A f(x) d\nu(x).$$

We think of the integrand $f(x)d\nu(x)$ as representing the “infinitesimal probability” that $X = x$.

When the state space $E = \mathbb{R}$, one furthermore has access to some statistical quantities related to X . The *expected value* (or *mean*) of X , denoted $\mathbb{E}X$, is the average value of X over Ω , weighted according to underlying probability measure $\mathbb{P}$:

$$\mathbb{E}X := \frac{1}{\mathbb{P}(\Omega)} \int_{\Omega} X(\omega) d\mathbb{P}(\omega) = \int_{\Omega} X(\omega) d\mathbb{P}(\omega).$$

Alternatively, by making the change of variables $x = X(\omega)$, we can also phrase this in terms of the distribution of X :

$$\mathbb{E}X = \int_{\mathbb{R}} x d\mathbb{P}_X(x).$$

The *variance* of X , denoted $\text{Var}(X)$, is defined as

$$\text{Var}(X) := \mathbb{E}(X - \mathbb{E}X)^2$$

and can be thought of as a measure of how concentrated X is around its mean.

If Y is another random variable defined on the same probability space as X , we say that X and Y are *independent* if for all $A, B \in \mathcal{F}$

$$\mathbb{P}(X \in A, Y \in B) = \mathbb{P}(X \in A)\mathbb{P}(Y \in B).$$

Intuitively, independence means that the random variables X and Y “do not contain any information about one another.” Put differently, knowing something about X (for instance that $X \in A$) tells one nothing about the likelihood that $Y \in B$. Under this naïve interpretation of independence, the definition follows quite naturally from the classic Bayesian identity

$$\mathbb{P}(X \in A, Y \in B) = \mathbb{P}(X \in A)\mathbb{P}(Y \in B \mid X \in A).$$

Finally, we say that X and Y are *identically distributed* if they have the same law.

2.2. Gaussian Random Variables. The *normal distribution* with mean μ and variance σ^2 , denoted $\mathcal{N}(\mu, \sigma^2)$, is the probability measure on $(\mathbb{R}, \mathcal{B})$ with Lebesgue-density

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/2\sigma^2}.$$

A real-valued random variable $X : (\Omega, \mathcal{F}, \mathbb{P}) \rightarrow (\mathbb{R}, \mathcal{B})$ is said to be *Gaussian* if there exist real numbers $\mu, \sigma \in \mathbb{R}$ such that $X \sim \mathcal{N}(\mu, \sigma^2)$, i.e. X follows a normal distribution with mean μ and variance σ^2 . In this case, we have that

$$\mathbb{E}X = \mu \quad \text{and} \quad \text{Var}(X) = \sigma^2$$

hence the choice of terminology for μ and σ^2 . A Gaussian random variable X is said to be *standard* if $\mu = 0$ and $\sigma^2 = 1$. One of the many things that makes Gaussian random variables special is the following:

Proposition 2.2.1 (Stability of the Normal Distribution). *The sum of independent Gaussians is again Gaussian. In particular, if $X \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $Y \sim \mathcal{N}(\mu_2, \sigma_2^2)$, then $X + Y \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$.*

Motivated by this fact, we say that a complex-valued random variable $Z : (\Omega, \mathcal{F}, \mathbb{P}) \rightarrow \mathbb{C}$ is a *standard complex Gaussian* if it can be written in the form $Z = X + iY$ for independent Gaussians X and Y with $X, Y \sim \mathcal{N}(0, \frac{1}{2})$. The stability of the normal distribution is moreover the key symmetry behind the most important theorem in probability:

Theorem 2.2.1 (Central Limit Theorem). *If $X_1, X_2, \dots$ are i.i.d. random variables with zero mean and unit variance, then*

$$S_n := \frac{X_1 + \dots + X_n}{\sqrt{n}} \rightarrow \mathcal{N}(0, 1)$$

in the sense of weak convergence of laws, that is, for all continuous bounded functions f , we have that

$$\mathbb{E}f(S_n) \rightarrow \mathbb{E}f(Z)$$

where $Z \sim \mathcal{N}(0, 1)$.

3. THE RANDOM DATA NLS

We are now in a position to study the periodic nonlinear Schrödinger equation with *randomized initial data*:

$$(RNLS) \quad \begin{cases} iu_t + \Delta u = \mathcal{N}_p(u) \\ u(0, x) = f(x, \omega) \end{cases}, \quad p \in 2\mathbb{N} + 1$$

There are two things to note about this equation that are different from the deterministic case. The first (and most subtle) is the non-linearity, which we have opted to write as $\mathcal{N}_p(u)$. This is because (as we shall see later) for some low-regularity choices of the initial data, the standard non-linearity $|u|^{p-1}u$ is ill-defined and hence the equation cannot be made sense of. In these cases, we thus need to “regularize” the nonlinearity so as to address this issue, which one does via *Wick ordering*; we discuss this in more detail later.

The second, of course, is the initial data, which we obtain in the obvious way: namely, by taking a random Fourier series in which the coefficients are i.i.d. standard complex Gaussians g_k (all defined on some fixed probability space $(\Omega, \mathcal{F}, \mathbb{P})$). We furthermore do one other thing, which is normalize these Gaussians so as to ensure that f belongs to some Sobolev space. In summary, our random initial data f is given by

$$f(x, \omega) = \sum_{k \in \mathbb{Z}^d} \frac{g_k(\omega)}{\langle k \rangle^\alpha} e^{ik \cdot x}, \quad \alpha = s + \frac{d}{2}.$$

Let’s take a moment to understand this initial data. Firstly, since

$$\|f\|_{H^{s'}}^2 = \sum_{k \in \mathbb{Z}^d} \langle k \rangle^{2s' - 2\alpha} |g_k|^2 = \sum_{k \in \mathbb{Z}^d} \langle k \rangle^{-d - 2(s - s')} |g_k|^2$$

and the series on the right is a.s. finite whenever $2(s - s') > 0$, we have that

$$f \in H^{s-}(\mathbb{T}^d) := \bigcap_{\varepsilon > 0} H^{s-\varepsilon}(\mathbb{T}^d) \quad \text{a.s.}$$

On face, this might seem strange since such an f does not necessarily belong to H^s , yet we have control over the regularity of f via the normalization factors $\langle k \rangle^\alpha$. The reason for choosing α as we did, and not with some small perturbation to ensure that $f \in H^s$, is that this would (by the same argument above) result in f belonging to $H^{s+\delta}$ for all sufficiently small δ , which is more regularity than we want. So, in some sense, our choice of α gets one as close to H^s as possible.

3.1. Probabilistic vs. Deterministic Scaling. It is reasonable to expect the critical regularity exponent in the case of random initial data to be different from that of the deterministic case. Here, we give a heuristic that demonstrates that they are indeed unequal and furthermore suggests a value for the critical exponent in the probabilistic setting. Recall that by a solution to (NLS) we really mean a solution to the integral equation

$$u(t) = e^{it\Delta}u_0 - i \int_0^t e^{i(t-s)\Delta} \mathcal{N}_p(u(s)) ds.$$

Formally, since u occurs on both sides of the above equation, we can build the solution u by iterating this identity. Moreover, intuitively, if we want the solution to belong to (say) H^s , then at each iteration we should *stay* in H^s . As a result, to have any hope at all of local well-posedness in H^s , it is necessary that the first iteration

$$u^{(1)}(t) = \int_0^t e^{i(t-s)\Delta} \mathcal{N}_p(e^{is\Delta}u_0) ds$$

have finite H^s norm. To understand the restrictions (if any) this imposes upon s , we consider the simple case in which the initial data is of a single large frequency $\sim N$, and then estimate the H^s norm of $u^{(1)}$ via its Fourier coefficients.

Let us first do this in the deterministic setting, for the purposes of both review and motivation. In this case, we can take, for instance,

$$u_0 = N^{-\alpha} \sum_{|k| \sim N} e^{ik \cdot x}, \quad \alpha = s + \frac{d}{2}$$

where the normalization by N^α is just so that $\|u_0\|_{H^s} \sim 1$. By writing $\mathcal{N}_p(u) = (u\bar{u})^{\frac{p-1}{2}}u$ (for the purposes of this heuristic we assume the standard nonlinearity) and Fourier expanding u , one easily checks that the Fourier coefficients $(\mathcal{N}_p(u))_k$ of the nonlinearity are given by

$$(\mathcal{N}_p(u))_k = \sum_{k_1 + \dots + k_p = k} u_{k_1} \overline{u_{k_2}} \dots \overline{u_{k_{p-1}}} u_{k_p}.$$

We then have that

$$\begin{aligned} (\mathcal{N}_p(e^{is\Delta}u_0))_k &= \sum_{k_1 + \dots + k_p = k} (e^{is\Delta}u_0)_{k_1} \overline{(e^{is\Delta}u_0)_{k_2}} \dots \overline{(e^{is\Delta}u_0)_{k_{p-1}}} (e^{is\Delta}u_0)_{k_p} \\ &= \sum_{k_1 + \dots + k_p = k} e^{-is|k_1|^2} (u_0)_{k_1} \cdot e^{is|k_2|^2} \overline{(u_0)_{k_2}} \dots e^{-is|k_p|^2} (u_0)_{k_p} \\ &= N^{-p\alpha} \sum_{\substack{|k_j| \sim N \\ k_1 + \dots + k_p = k}} e^{is(-|k_1|^2 + |k_2|^2 - \dots - |k_p|^2)} \end{aligned}$$

and hence

$$(u^{(1)})_k = \int_0^t e^{i(s-t)|k|^2} (\mathcal{N}_p(e^{is\Delta}u_0))_k ds = N^{-p\alpha} \sum_{\substack{|k_j| \sim N \\ k_1 + \dots + k_p = k}} e^{-it|k|^2} \int_0^t e^{is\Omega} ds$$

where $\Omega = |k|^2 - |k_1|^2 + |k_2|^2 - \dots - |k_p|^2$. In particular, we see that

$$(u^{(1)})_k \sim N^{-p\alpha} \sum_{\substack{|k_j| \sim N \\ k_1 + \dots + k_p = k}} \frac{1}{\langle \Omega \rangle}.$$

Assuming $\Omega = 0$ for simplicity, a dimension counting argument on the sum then yields

$$(u^{(1)})_k \sim N^{-p\alpha} \sum_{\substack{|k_j| \sim N \\ k_1 + \dots + k_p = k}} \sim N^{-p\alpha + pd - d - 2}.$$

From this, we deduce that

$$\|u^{(1)}\|_{H^s} = \left(\sum_{k \in \mathbb{Z}^d} (1 + |k|^2)^s |(u^{(1)})_k|^2 \right)^{1/2} \sim N^{-p\alpha + pd - d - 2 + \alpha}$$

and hence

$$\|u^{(1)}\|_{H^s} \lesssim 1 \iff \alpha \geq d - \frac{2}{p-1} \iff s \geq \frac{d}{2} - \frac{2}{p-1} := s_{\text{cr}}.$$

Indeed, in the deterministic setting, (NLS) is locally well-posed for $s > s_{\text{cr}}$ and ill-posed for $s < s_{\text{cr}}$.

We now apply the same argument in the random setting. In this case, our initial data may look something like

$$u_0 = N^{-\alpha} \sum_{|k| \sim N} g_k e^{ik \cdot x},$$

where the g_k are i.i.d. standard complex Gaussians. Doing everything exactly as before, we find this time that

$$(u^{(1)})_k \sim N^{-p\alpha} \sum_{\substack{|k_j| \sim N \\ k_1 + \dots + k_p = k}} \frac{1}{\langle \Omega \rangle} g_{k_1} \overline{g_{k_2}} \dots \overline{g_{k_{p-1}}} g_{k_p} \stackrel{\Omega=0}{\sim} N^{-p\alpha} \sum_{\substack{|k_j| \sim N \\ k_1 + \dots + k_p = k}} g_{k_1} \overline{g_{k_2}} \dots \overline{g_{k_{p-1}}} g_{k_p}.$$

where again we assume $\Omega = 0$ for simplicity. If we furthermore assume that there are no pairings among the g_{k_j} , then the summands above are independent and subsequently with high probability the sum has size $\sim N^{(pd-d-2)/2}$. The factor of $\frac{1}{2}$ comes from square-root cancellation properties of sums of independent random variables. For instance, recall that the Central Limit Theorem says that for i.i.d. centered and normalized random variables $X_1, \dots, X_m$ we have that, for m large,

$$\frac{X_1 + \dots + X_m}{\sqrt{m}} \approx \mathcal{N}(0, 1) \quad \text{and hence, with high probability, } X_1 + \dots + X_m \sim \sqrt{m}.$$

In our case, all we have done is applied this idea (although in a more general case) with $m \sim N^{pd-d-2}$. In particular, it follows that in the probabilistic case,

$$\begin{aligned} \|u^{(1)}\|_{H^s} \lesssim 1 &\iff -p\alpha + \frac{1}{2}(pd - d - 2) + \alpha \leq 0 \\ &\iff (p-1)\alpha \geq \frac{1}{2}(p-1)d - 1 \\ &\iff \alpha \geq \frac{d}{2} - \frac{1}{p-1} \\ &\iff s \geq -\frac{1}{p-1} := s_{\text{pr}}. \end{aligned}$$

In tandem with the deterministic case, it is therefore plausible that we may have almost sure local well-posedness in the *probabilistically subcritical regime* $s > s_{\text{pr}}$. In the following section, we'll see that this is indeed the case.

3.2. Well-Posedness Theory. We conclude the talk by giving a very rough overview of the current well-posedness theory of (RNLS). Something interesting to note is that s_{pr} is *negative*, meaning that for some probabilistically subcritical cases (namely, $s_{\text{pr}} < s < 0$) we are operating in “rough” Sobolev spaces that contain highly irregular objects that needn't even be functions. In this case, the nonlinearity $|u|^{p-1}u$ becomes problematic as nonlinear operations don't make sense when applied to distributions. To remedy this, we replace $|u|^{p-1}u$ with its so-called *Wick-ordering*. We do not go through the trouble of defining Wick-orderings precisely (as they will not be relevant to us; for a detailed treatment, see [DNY24]), however the basic idea is that by truncating the Fourier series of u to include only the first $\sim N$ nodes, we obtain a sequence (u_N) of smooth approximations to u in which the nonlinearities $|u_N|^{p-1}u_N$ make sense and can be thought of as approximations to the nonsensical object $|u|^{p-1}u$. By adding some regularizing terms to each $|u_N|^{p-1}u_N$, we can ensure that the sequence converges to a non-trivial and well-defined limit, which we call the *Wick-ordering* of $|u|^{p-1}u$ and denote by $W^p(u)$. Our choice of nonlinearity in (RNLS) can thus be summarized as

$$\mathcal{N}_p(u) = \begin{cases} |u|^{p-1}u & \text{if } s \geq 0 \\ W^p(u) = \text{“version of } |u|^{p-1}u \text{ that makes sense”} & \text{if } s < 0 \end{cases}$$

(in fact, the Wick-ordered NLS is equivalent to the standard one when $s \geq 0$). With this (important) point out of the way, we have the following result providing a complete picture of the local theory in the probabilistically subcritical regime:

Theorem 3.2.1 (Deng-Nahmod-Yue, 2020 [DNY20]). *Let $p \geq 3$ be odd, $s > s_{\text{pr}}$, and $\alpha = s + \frac{d}{2}$. Then, almost surely, there exists a local strong H_x^{s-} solution to (RNLS) that is unique in a suitable sense.*

We have intentionally been a bit vague as the true result is a little technical. By uniqueness in a “suitable sense”, we mean that the solution is the unique limit among all “canonical finite-dimensional truncations” of the system (see [DNY20] for details). Deng, Nahmod, and Yue furthermore conjecture that in the probabilistically supercritical regime $s < s_{\text{pr}}$ (RNLS) is almost surely *ill-posed*, however they remark that proving this seems to be out of the reach of their current methods and subsequently this part of the analysis remains open.

The next question we wish to address is under what conditions we can extend the local solutions of the subcritical regime to global ones. In order to have any hope of doing this, one needs to understand how randomness propagates under the flow of (RNLS). This is in stark contrast with the local theory, in which case we can get away with little to no understanding of this on the basis that the nonlinearity shouldn’t have too large an effect on the distribution of the initial data due to the small time scales. The key to understanding this propagation of randomness is the *Gibbs measure*.

Formally, the *Gibbs measure* is a probability measure μ on the space $\mathcal{S}'(\mathbb{T}^d)$ of distributions on $\mathbb{T}^d$ defined by the relation

$$d\mu(u) \sim e^{-\mathcal{H}[u]} \prod_{x \in \mathbb{T}^d} du(x).$$

Here, we think of the product $\prod_{x \in \mathbb{T}^d} du(x)$ as the “Lebesgue measure” on the function space $\mathbb{R}^{\mathbb{T}^d}$, and $\mathcal{H}$ denotes the *Hamiltonian* functional of the system,

$$\mathcal{H}[u](t) := \frac{1}{2} \int_{\mathbb{T}^d} |\nabla u(t, x)|^2 dx + V[u](t),$$

which is a conserved quantity given by a kinetic energy term and potential energy term $V[u](t)$ which depends on the choice of nonlinearity. In a few select cases, the Gibbs measure can be made sense of rigorously even though the Lebesgue measure factor cannot. Specifically,

- for $d = 1, 2$ it can be constructed for any p ;
- for $d = 3$ it can be constructed for $p = 3$ (for other p the construction is either unknown or has been proven impossible); and,
- for $d \geq 4$ its construction is unknown or has been proven impossible for every p .

In general, constructing the Gibbs measure rigorously and understanding its properties under various dynamics is a major problem in QFT.

The upshot is that, when defined, the Gibbs measure is *invariant* under the flow of (RNLS). Roughly, this allows to estimate the statistics of solutions at larger time scales since, intuitively, we expect the randomness in the system to converge to the equilibrium state (the Gibbs measure) as $t \rightarrow \infty$. In this sense, the invariance of the Gibbs measure plays an entirely analogous role to the conserved quantities we saw in class in allowing us to control the growth of the local-in-time solutions and extend them to global solutions. This was first realized by Bourgain in the ’90s, who showed the following:

Theorem 3.2.2 (Bourgain, 1994, 1996 [Bou94; Bou96]). *Suppose that $d = 1$ and $p \geq 3$ or $(d, p) = (2, 3)$. Then, for $\alpha = 1$, (RNLS) is almost surely globally well-posed.*

The picture in the 2D was recently completed by Deng, Nahmod, and Yue:

Theorem 3.2.3 (Deng-Nahmod-Yue, 2024 [DNY24]). *Suppose that $d = 2$ and $p \geq 5$. Then, for $\alpha = 1$, (RNLS) is almost surely globally well-posed.*

The critical case $(d, p) = (3, 3)$ is very hard and remains open.

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